WORKED SOLUTION

STEP I 2017, Question 3

Pure • Tangents to a Parabola

STEP I 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q3

The key idea

Everything here can be written in terms of the parameters p and q, so the plan is to find the coordinates of R, S and T and then compute both areas directly. Triangle RST is especially friendly because its side ST lies along the y-axis: its height is simply the distance of R from that axis. For triangle OPQ, with a vertex at the origin, the formula $\frac{1}{2}|x_1y_2 - x_2y_1|$ does the work in one line.

Solution

The tangent at P

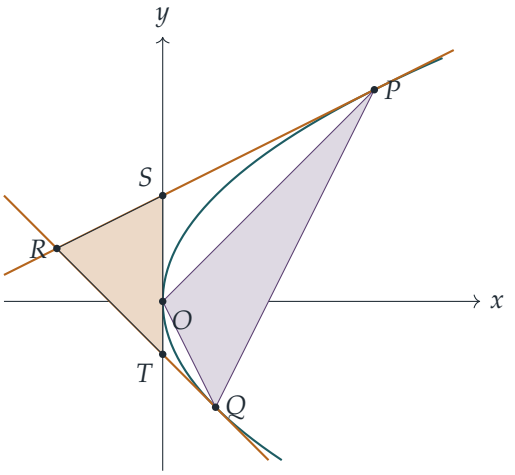
Differentiating $y^2 = 4ax$ implicitly gives $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4a$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{y}$. At P, where $y = 2ap \neq 0$ (since $a, p > 0$), the gradient is

$$\frac{2a}{2ap} = \frac{1}{p}.$$

The tangent through $P(ap^2, 2ap)$ is therefore

$$y - 2ap = \frac{1}{p}\left(x - ap^2\right) \quad \Longrightarrow \quad y = \frac{1}{p}x - ap + 2ap = \frac{1}{p}x + ap,$$

as required. In the same way, since Q has the same form with q in place of p (and $q \neq 0$), the tangent at Q is $y = \dfrac{1}{q}x + aq$.



The sketch shows the case $a = 1$, $p = 2$, $q = -1$: R lies to the left of the y-axis, between S and T in height.

The points R, S and T

Putting $x = 0$ in each tangent gives

$$S = (0,\, ap), \qquad T = (0,\, aq).$$

At R the two tangents meet, so

$$\frac{x}{p} + ap = \frac{x}{q} + aq \quad \Longrightarrow \quad x\left(\frac{1}{p} - \frac{1}{q}\right) = a(q-p) \quad \Longrightarrow \quad x \cdot \frac{q-p}{pq} = a(q-p).$$

Since $p > 0 > q$ we have $q - p \neq 0$, so we may divide by it (this is also why the tangents are not parallel and R exists). Hence $x = apq$, and then $y = \dfrac{apq}{p} + ap = a(p+q)$:

$$R = (apq,\ a(p+q)).$$

Ask yourself Which side of the y-axis is R on, and why does it matter for the area?

Because $p > 0$ and $q < 0$, the product pq is negative, so the x-coordinate apq of R is negative: R is to the left of the axis, at distance $a|pq| = -apq$ from it. Areas must be positive, so we keep track of signs rather than writing apq as a "height".

Area of triangle RST

Take ST as the base. It lies along the y-axis with length $ap - aq = a(p-q)$, which is positive since $p > q$. The perpendicular height from R to the y-axis is $-apq$. So

$$[RST] = \tfrac{1}{2} \cdot a(p-q) \cdot (-apq) = -\tfrac{1}{2}a^2pq(p-q).$$

Area of triangle OPQ

For a triangle with vertices O, (x_1, y_1) and (x_2, y_2), the area is $\frac{1}{2}|x_1y_2 - x_2y_1|$. With $P(ap^2, 2ap)$ and $Q(aq^2, 2aq)$,

$$x_1y_2 - x_2y_1 = ap^2 \cdot 2aq - aq^2 \cdot 2ap = 2a^2pq(p-q).$$

Here $a^2 > 0$, $pq < 0$ and $p - q > 0$, so this is negative, and

$$[OPQ] = \tfrac{1}{2} \cdot (-2a^2pq(p-q)) = -a^2pq(p-q).$$

Conclusion

Comparing the two results,

$$[OPQ] = -a^2pq(p-q) = 2\left(-\tfrac{1}{2}a^2pq(p-q)\right) = 2\,[RST],$$

so the area of triangle OPQ is twice the area of triangle RST, as required.

Checks

In the sketched case $a = 1$, $p = 2$, $q = -1$ we get $P(4,4)$, $Q(1,-2)$, $R(-2,1)$, $S(0,2)$ and $T(0,-1)$. Then $[RST] = \frac{1}{2} \cdot 3 \cdot 2 = 3$ and $[OPQ] = \frac{1}{2}|4 \cdot (-2) - 1 \cdot 4| = 6$, twice as big. You can also check that R satisfies both tangent equations: $1 = \frac{-2}{2} + 2$ and $1 = \frac{-2}{-1} - 1$. The identity was also confirmed by computer, in exact arithmetic, for 2000 random choices of a, p and q.

Common slips

- Writing the height of triangle RST as apq: this is negative, because R is on the far side of the y-axis from the curve.
- Dropping the modulus in $\frac{1}{2}|x_1y_2 - x_2y_1|$ and ending up with a negative area for OPQ.
- Dividing by $q - p$ without saying why it is non-zero.

Beyond STEP

The parametrisation $(at^2, 2at)$ makes every point of the parabola correspond to one number t, and the tangents at $t = p$ and $t = q$ always meet at $(apq, a(p+q))$: the "product" and "sum" of the parameters. Results like this, together with the $2:1$ area ratio here, are part of the classical projective geometry of conics, where tangents and chords are studied using parameters and determinants. The determinant formula for the area of a triangle is the starting point for seeing area as a signed, linear quantity, which is developed in Strang, *Introduction to Linear Algebra*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with parametric curves.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.